

DJELLEL DIFALLAH

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RESEARCH INTERESTS

My research combines data and knowledge engineering with machine learning to address challenges arising from incomplete and heterogeneous data. My work spans knowledge representation and completion, question answering and graph reasoning, multimodal learning, and explanation of model predictions. I place particular emphasis on developing interpretable reasoning systems that can support human-machine collaboration for knowledge completion and curation.

EDUCATION

Ph.D. in Computer Science University of Fribourg, Switzerland Thesis: “ <i>Quality of Service in Crowd-Powered Systems</i> ” Advisor: Prof. Philippe Cudré-Mauroux	May 2011 – Jun 2015
M.S. in Computer Science (Fulbright Scholarship) University of Louisiana at Lafayette, USA Project: “ <i>Frequent Association Action Rules Mining Using FP-Trees</i> ” Advisors: Prof. Vijay V. Raghavan and Prof. Ryan Benton	Aug 2009 – May 2011
Dipl.-Ing. in Informatics University of Science and Technology Houari Boumediene, Algeria	Oct 1999 – Dec 2004

ACADEMIC APPOINTMENTS AND PROFESSIONAL EXPERIENCE

Computer Science Program Head , NYU Abu Dhabi	Sep 2025 – Present
Global Network Assistant Professor of Computer Science (Courant Institute of Mathematical Sciences, NYU); Assistant Professor of Computer Science (NYU Abu Dhabi)	Sep 2020 – Present
Research Scientist Wikimedia Foundation, Switzerland	Oct 2019 – Aug 2020
Adjunct Assistant Professor Stern School of Business, New York University, USA	Spring 2019
Postdoctoral Associate (Data Science Fellow) Center for Data Science, New York University, USA	May 2017 – Oct 2019
Senior Research Scientist eXascale Infolab, University of Fribourg, Switzerland	Oct 2015 – May 2017
Research Assistant eXascale Infolab, University of Fribourg, Switzerland	May 2011 – Jul 2015
Intern Microsoft Research, Cloud and Information Services Lab, Mountain View, CA, USA	Summer 2013
Intern (Google Summer of Code) Mentoring Organization: DrizzleDB Project	Summer 2010
Information Management Engineer Schlumberger Information Solutions, Algeria	2006 – 2009
Engineer EEPAD Internet Services Provider, Algeria	2005 – 2006

TEACHING EXPERIENCE

New York University Abu Dhabi – Graduate Courses (Instructor)

- Fall 2026 – **Introduction to Data Science** (IDSC-GH 5010)
Master of Science in Interdisciplinary Data Science and AI; enrollment: 20 students.

New York University Abu Dhabi – Undergraduate Courses (Instructor)

- Spring 2026 – **Research Seminar** (CS-UH 3090)
Enrollment: 77 students.
- Fall 2025 – **Data Structures** (CS-UH 1050)
Enrollment: 24 students.
- Spring 2025 – **Data Structures** (CS-UH 1050, Section 001)
Enrollment: 23 students.
- Spring 2025 – **Directed Study: Graph Machine Learning** (CS-UH 3250)
Enrollment: 6 students.
- Fall 2024 – **Big Data Systems** (CS-UH 3260)
Special Topics in Computer Science; enrollment: 24 students.
- Fall 2023 – **Data Structures** (CS-UH 1050, Section 002)
Enrollment: 24 students.
- Spring 2023 – **Big Data Systems** (CS-UH 3260)
Special Topics in Computer Science; enrollment: 24 students.
- Fall 2022 – **Discrete Mathematics** (CS-UH 1002, Section 004)
Enrollment: 23 students.
- Spring 2022 – **Directed Study: Probabilistic Machine Learning** (CS-UH 3250)
Enrollment: 1 student.
- Fall 2021 – **Discrete Mathematics** (CS-UH 1002, Section 001)
Enrollment: 24 students.
- Fall 2021 – **Discrete Mathematics** (CS-UH 1002, Section 002)
Enrollment: 24 students.
- Spring 2021 – **Data Structures** (CS-UH 1050, Section 003)
Enrollment: 24 students.

Stern School of Business, NYU – Instructor

- Spring 2019 – **Data Mining for Business Analytics (MBA)** (INFO-GB.3336.20)
Enrollment: 21 students.

University of Fribourg

- Fall 2016 – **Big Data Infrastructures** (Master's)
Co-instructor with Prof. Philippe Cudré-Mauroux.
- Fall 2015 – **Data Science Seminar** (Master's)
Co-instructor with Dr. Mourad Khayati.
- 2011 – 2014 – **Social Computing** (Master's)
Teaching assistant; instructor: Prof. Philippe Cudré-Mauroux.

AWARDS AND HONORS

Best In-Use Paper Award, ISWC	2025
<i>Open Government Data as Multi-dimensional 5 Star Data: cube.link</i>	
With Michael Luggen, Benedikt Hitz, Julien Audiffren, Jean-Luc Cochard, and Philippe Cudré-Mauroux	
NYU Abu Dhabi Palm Teaching Award	AY 2024 – 2025
Honorarium: USD 1,000	
Distinguished Reviewers Board, ACM Transactions on the Web	2022
Outstanding Reviewer, The Web Conference	2018
Best Computer Science Ph.D. Thesis Award, University of Fribourg	Feb 2016
Awarded: CHF 3,000	
Research Visit Grant, European Science Foundation	Spring 2015
Awarded: EUR 1,350	
Travel Grant: HCOMP Doctoral Consortium	Oct 2014
Awarded: USD 700	
University of Louisiana at Lafayette Honors (GPA: 4.0)	2011

GRANTS AND FUNDED PROJECTS

Soft Power vs. Material Power: Comparing the Effects of Chinese Media Campaigns and Overseas Investment on Local Perceptions

Awardees: Sichen Li (LSE), Hao Zhang (NYU NY), and Djellel Difallah (NYUAD)

Role: PI leading the work package developing a temporal knowledge graph from GDELT to support longitudinal analysis of Chinese media campaigns, overseas investment, and local perceptions.

Funding source: LSE–NYU Research Seed Fund

Grant period: 1 Aug 2026 – 31 Jan 2028

Grant value: USD 13,000

CityGraph: A Participatory Open Data Framework

Role: PI, with Prof. Etienne Wasmer (NYUAD Social Science)

Funding source: Center for Interacting Urban Networks, NYUAD

Grant period: 2022 – 2024

Grant value: 2-year postdoctoral position, 1 month of summer salary, USD 5,000 research funds

FashionBrain: Understanding Europe's Fashion Data Universe

Role: Proposal development and project launch (involved through May 2017, prior to joining NYU).

Funding source: European Project Horizon 2020 (Big Data PPP)

Grant period: 1 Jan 2017 – 31 Dec 2019

Grant value: EUR 2.9 million total; University of Fribourg's share: EUR 690,000

Query Expansion: Deep Learning and Crowdsourcing

Role: Key personnel and lead grant writer

Funding source: Swiss Commission for Technology and Innovation (CTI)

Grant period: 2016

Grant value: CHF 210,000

PUBLICATIONS

Google Scholar metrics (October 2026):

Citations: 3,966

H-index: 21

I10-index: 27

[†] Direct supervision.

Refereed Publications

- AACL- Prince Larbi Ampofo[†], Ryoji Kubo[†], and **Djellel Difallah**. DBRAG: Multi-Table IJCNLP '26 Retrieval-Augmented Generation for Complex Database Queries. In *AACL-IJCNLP*, 2026. (Accepted, main conference). [\[Preprint\]](#)
- AACL- Ola El Khatib[†] and **Djellel Difallah**. Explore-over-Graph: Hybrid Embedding-LLM IJCNLP '26 Reasoning for Knowledge Graph Question Answering under Incompleteness. In *Findings of AACL-IJCNLP*, 2026. (Accepted, Findings). [\[Preprint\]](#)
- LoG '26 Ryoji Kubo[†], Samyam Lamichhane[†], and **Djellel Difallah**. RAW-Explainer: Post-hoc Explanations of Path Reasoning Graph Neural Networks for Knowledge Graph Completion. In *Learning on Graphs Conference (LoG)*, 2026. (Accepted). [\[Preprint\]](#)
- IJCAI '26 Yue Wang, **Djellel Difallah**, Areg Karapetyan, and Samer Madanat. UniST-Pred: A Robust Unified Framework for Spatio-Temporal Traffic Forecasting in Transportation Networks Under Disruptions. In *Proceedings of the Thirty-Fifth International Joint Conference on Artificial Intelligence (IJCAI-26)*, pages 7501–7511, 2026.
- KDD '26 Dongzhe Fan, Jiajin Liu, Yi Fang, **Djellel Difallah**, and Qiaoyu Tan. MLaGA: Multimodal Large Language and Graph Assistant. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26)*, pages 1027–1038, Jeju Island, Republic of Korea, 2026. ACM.
- SIGIR '26 Rohan Sabu[†], Ola El Khatib[†], and **Djellel Difallah**. A Demonstration of WikiRAG: An Evidence-Based Link Prediction for Wikidata with Retrieval Augmented Generation. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '26, pages 5204–5208. ACM, 2026. [\[Live demo\]](#)
- KDD '25 **Djellel Difallah**. WikiRAG: Revisiting Wikidata KGC Datasets with Community Updates and Retrieval-Augmented Generation. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '25)*, pages 5391–5401, 2025.
- ISWC '25 Michael Luggen, Benedikt Hitz, Julien Audiffren, **Djellel Difallah**, Jean-Luc Cochard, and Philippe Cudré-Mauroux. Open Government Data as Multi-dimensional 5 Star Data: cube.link. In *ISWC (2)*, Lecture Notes in Computer Science, pages 363–378. Springer, 2025. [\(Best In-Use Paper Award\)](#)
- SDM '24 Ryoji Kubo[†] and **Djellel Difallah**. XGExplainer: Robust Evaluation-based Explanation for Graph Neural Networks. In *Proceedings of the 2024 SIAM International Conference on Data Mining (SDM)*, pages 64–72. SIAM, 2024.
- PVLDB '24 Luca Althaus, Mourad Khayati, Abdelouahab Khelifati, Anton Dignös, **Djellel Difallah**, and Philippe Cudré-Mauroux. SEER: An End-to-End Toolkit for Benchmarking Time Series Database Systems in Monitoring Applications. *Proceedings of the VLDB Endowment*, 17(12): 4361–4364, 2024. [\[Live demo\]](#)

- SIGIR '23 Farah Atif[†], Ola El Khatib[†], and **Djellel Difallah**. BeamQA: Multi-hop knowledge graph question answering with sequence-to-sequence prediction and beam search. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 781–790, 2023.
- PVLDB '23 Abdelouahab Khelifati, Mourad Khayati, Anton Dignös, **Djellel Difallah**, and Philippe Cudré-Mauroux. TSM-Bench: Benchmarking Time Series Database Systems for Monitoring Applications. *Proceedings of the VLDB Endowment*, 16(11):3363–3376, 2023.
- ICDM '22 Nadya Abdel Madjid[†], Ola El Khatib[†], Shuang Gao[†], and **Djellel Difallah**. HyperKGQA: Question Answering over Knowledge Graphs using Hyperbolic Representation Learning. In *IEEE International Conference on Data Mining (ICDM)*, pages 309–318. IEEE, 2022.
- Big Data '22 Natalia Ostapuk, **Djellel Difallah**, and Philippe Cudré-Mauroux. ParaGraph: Mapping Wikidata Tail Entities to Wikipedia Paragraphs. In *IEEE Big Data*, pages 6008–6017. IEEE, 2022.
- Big Data '22 **Djellel Difallah**, Diego Sáez-Trumper, Eriq Augustine, Robert West, and Leila Zia. Crosslingual Section Title Alignment in Wikipedia. In *IEEE Big Data*, pages 5892–5901. IEEE, 2022.
- WWW '21 Michael Luggen, Julien Audiffren, **Djellel Difallah**, and Philippe Cudré-Mauroux. Wiki2Prop: A multimodal approach for predicting Wikidata properties from Wikipedia. In *Proceedings of the Web Conference 2021*, pages 2357–2366, 2021.
- CIKM '21 Martin Gerlach, Marshall Miller, Rita Ho, Kosta Harlan, and **Djellel Difallah**. Multilingual Entity Linking System for Wikipedia with a Machine-in-the-Loop Approach. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management (CIKM)*, pages 3818–3827, 2021.
- CIKM '20 Lei Han, Alessandro Checco, **Djellel Difallah**, Gianluca Demartini, and Shazia Sadiq. Modelling user behavior dynamics with embeddings. In *Proceedings of the 29th ACM International Conference on Information & Knowledge Management (CIKM)*, pages 445–454, 2020.
- ISWC '19 Michael Luggen, **Djellel Difallah**, Cristina Sarasua, Gianluca Demartini, and Philippe Cudré-Mauroux. Non-parametric Class Completeness Estimators for Collaborative Knowledge Graphs - The Case of Wikidata. In *ISWC (1)*, volume 11778 of *Lecture Notes in Computer Science*, pages 453–469. Springer, 2019.
- TSC '19 **Djellel Difallah**, Alessandro Checco, Gianluca Demartini, and Philippe Cudré-Mauroux. Deadline-Aware Fair Scheduling for Multi-Tenant Crowd-Powered Systems. *ACM Transactions on Social Computing*, 2(1):3:1–3:29, 2019.
- JCSCW '19 Cristina Sarasua, Alessandro Checco, Gianluca Demartini, **Djellel Difallah**, Michael Feldman, and Lydia Pintscher. The Evolution of Power and Standard Wikidata Editors: Comparing Editing Behavior over Time to Predict Lifespan and Volume of Edits. *Computer Supported Cooperative Work (CSCW)*, 28(5):843–882, 2019.
- LREC '18 Julien Plu, Roman Prokofyev, Alberto Tonon, Philippe Cudré-Mauroux, **Djellel Difallah**, Raphaël Troncy, and Giuseppe Rizzo. Sanaphor++: Combining Deep Neural Networks with Semantics for Coreference Resolution. In *LREC*. European Language Resources Association (ELRA), 2018.
- WSDM '18 **Djellel Difallah**, Elena Filatova, and Panos Ipeirotis. Demographics and Dynamics of Mechanical Turk Workers. In *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining*, WSDM '18, pages 135–143, 2018.

- SEMANTiCS Roman Prokofyev, Michael Luggen, **Djellel Difallah**, and Philippe Cudré-Mauroux.
'17 SwissLink: High-Precision, Context-Free Entity Linking Exploiting Unambiguous Labels. In *Proceedings of the 13th International Conference on Semantic Systems*, pages 65–72. ACM, 2017.
- FnT Web '17 Gianluca Demartini, **Djellel Difallah**, Ujwal Gadiraju, and Michele Catasta. An Introduction to Hybrid Human-Machine Information Systems. *Foundations and Trends® in Web Science*, 7(1):1–87, 2017.
- WWW '16 **Djellel Difallah**, Gianluca Demartini, and Philippe Cudré-Mauroux. Scheduling Human Intelligence Tasks in Multi-Tenant Crowd-Powered Systems. In *WWW*, pages 855–865. ACM, 2016.
- ISWC '16 Alberto Tonon, Victor Felder, **Djellel Difallah**, and Philippe Cudré-Mauroux.
Resource VoldemortKG: Mapping schema.org and Web Entities to Linked Open Data. In *ISWC (2)*, volume 9982 of *Lecture Notes in Computer Science*, pages 220–228, 2016.
- ISWC '15 Roman Prokofyev, Alberto Tonon, Michael Luggen, Loic Vouilloz, **Djellel Difallah**, and Philippe Cudré-Mauroux. SANAPHOR: Ontology-Based Coreference Resolution. In *ISWC (1)*, volume 9366 of *Lecture Notes in Computer Science*, pages 458–473. Springer, 2015.
- WWW '15 **Djellel Difallah**, Michele Catasta, Gianluca Demartini, Panagiotis G. Ipeirotis, and Philippe Cudré-Mauroux. The Dynamics of Micro-Task Crowdsourcing: The Case of Amazon MTurk. In *WWW*, pages 238–247. ACM, 2015.
- SIGMOD '15 Dana Van Aken, **Djellel Difallah**, Andrew Pavlo, Carlo Curino, and Philippe
Demo Cudré-Mauroux. BenchPress: Dynamic Workload Control in the OLTP-Bench Testbed. In *SIGMOD Conference*, pages 1069–1073. ACM, 2015.
- SoCC '14 Carlo Curino, **Djellel Difallah**, Chris Douglas, Subru Krishnan, Raghu Ramakrishnan, and Sriram Rao. Reservation-based Scheduling: If You're Late Don't Blame Us! In *SoCC*, pages 2:1–2:14. ACM, 2014.
- HCOMP '14 **Djellel Difallah**, Michele Catasta, Gianluca Demartini, and Philippe Cudré-Mauroux. Scaling-Up the Crowd: Micro-Task Pricing Schemes for Worker Retention and Latency Improvement. In *HCOMP*, pages 50–58. AAAI, 2014.
- WWW '14 Michele Catasta, Alberto Tonon, **Djellel Difallah**, Gianluca Demartini, Karl Aberer, and Philippe Cudré-Mauroux. Hippocampus: answering memory queries using transactive search. In *WWW (Companion Volume)*, pages 535–540. ACM, 2014.
- PVLDB '14 Michele Catasta, Alberto Tonon, **Djellel Difallah**, Gianluca Demartini, Karl Aberer, and
Demo Philippe Cudré-Mauroux. TransactiveDB: Tapping into Collective Human Memories. *Proc. VLDB Endow.*, 7(14):1977–1980, 2014.
- WWW '13 **Djellel Difallah**, Gianluca Demartini, and Philippe Cudré-Mauroux. Pick-a-Crowd: tell me what you like, and I'll tell you what to do. In *WWW*, pages 367–374. International World Wide Web Conferences Steering Committee / ACM, 2013.
- PVLDB '13 **Djellel Difallah**, Andrew Pavlo, Carlo Curino, and Philippe Cudré-Mauroux. OLTP-Bench: An Extensible Testbed for Benchmarking Relational Databases. *Proc. VLDB Endow.*, 7(4): 277–288, 2013.
- VLDB J. '13 Gianluca Demartini, **Djellel Difallah**, and Philippe Cudré-Mauroux. Large-scale linked data integration using probabilistic reasoning and crowdsourcing. *The VLDB Journal*, 22(5): 665–687, 2013.

- IEEE IC '13 **Djellel Difallah**, Philippe Cudré-Mauroux, and Sean A. McKenna. Scalable Anomaly Detection for Smart City Infrastructure Networks. *IEEE Internet Computing*, 17(6):39–47, 2013.
- WWW '12 Gianluca Demartini, **Djellel Difallah**, and Philippe Cudré-Mauroux. ZenCrowd: leveraging probabilistic reasoning and crowdsourcing techniques for large-scale entity linking. In *WWW*, pages 469–478. ACM, 2012.

Workshop Papers

- Big Data '22 Lifan Yu[†], Nadya Abdel Madjid[†], and **Djellel Difallah**. CrunchQA: A Synthetic Dataset for Question Answering over Crunchbase Knowledge Graph. In *IEEE Big Data*, pages 4635–4641. IEEE, 2022.
- ISWC '20 Natalia Ostapuk, **Djellel Difallah**, and Philippe Cudré-Mauroux. SectionLinks: Mapping Orphan Wikidata Entities onto Wikipedia Sections. In *Wikidata@ISWC*, volume 2773 of *CEUR Workshop Proceedings*. CEUR-WS.org, 2020.
- CIKM '12 Carlo Curino, **Djellel Difallah**, Andrew Pavlo, and Philippe Cudré-Mauroux. Benchmarking OLTP/web databases in the cloud: the OLTP-bench framework. In *CloudDB@CIKM*, pages 17–20. ACM, 2012.
- WWW '12 **Djellel Difallah**, Gianluca Demartini, and Philippe Cudré-Mauroux. Mechanical Cheat: Spamming Schemes and Adversarial Techniques on Crowdsourcing Platforms. In *CrowdSearch*, volume 842 of *CEUR Workshop Proceedings*, pages 26–30. CEUR-WS.org, 2012.
- ICDM '11 **Djellel Difallah**, Ryan G. Benton, Vijay V. Raghavan, and Tom Johnsten. FAARM: Frequent Association Action Rules Mining Using FP-Tree. In *ICDM Workshops*, pages 398–404. IEEE Computer Society, 2011.

Under Review

- Quentin Nater, Zakhar Tymchenko, Philippe Cudré-Mauroux, **Djellel Difallah**, and Mourad Khayati. CleanImp: A Comprehensive Evaluation of Data Imputation Impact on Time Series Downstream Tasks.
- Ola El Khatib[†] and **Djellel Difallah**. QALLM: Path Generation and Knowledge Graph Embeddings for Multi-hop KGQA.

Thesis

- Ph.D. '15 **Djellel Difallah**. *Quality of Service in Crowd-powered Systems*. PhD thesis, University of Fribourg, Switzerland, 2015.

RESEARCH SUPERVISION

Ph.D. Students

- Ola El Khatib, NYU Tandon. Aug 2021 – Present. Expected graduation December 2026.

Postgraduate Researchers (Bachelor's Graduates)

- Samyam Lamichhane. Sep 2025 – Aug 2026. (Currently: Ph.D. student, University of Virginia)
- Ryoji Kubo. Sep 2024 – Aug 2025. (Currently: Ph.D. student, National University of Singapore)
- Prince Larbi Ampofo. Nov 2023 – Aug 2025. (Currently: master's student, Georgia State University)

Research Associates

- Wenqing Li. Sep 2023 – Aug 2024. (Currently: Hunan University, China)
- Rashid Alyassi. Jan 2023 – Aug 2023. (Currently: Ph.D. student, ETH Zurich)
- Farah Atif. Apr 2022 – Apr 2023. (Currently: Ph.D. student, MBZUAI)
- Nadya Abdel Madjid. Feb 2021 – Aug 2022. (Currently: Ph.D. student, Khalifa University)
- Bilal Saleem. Sep 2020 – Aug 2021. (Currently: Ph.D. student, Purdue University)

STUDENT ADVISING

Undergraduate Capstone Students at NYUAD

AY 2026 – 2027

- Viesturs Olins: *"Mapping China's Soft Power: A Longitudinal, Country-Specific Index Using Large-Scale News Data"*
- Dijar Selami: *"Graph-Based Join-Aware Table Selection for Multi-Table Question Answering"*
- Tinh Ngo: *"A Visual and Interactive Framework for Explainable Knowledge Graph Link Prediction"*

AY 2025 – 2026

- Reginald Appiah-Sekyere: *"A Hybrid Framework for RDF Querying: Integrating Vector-Based Search for Efficient Read Performance"*
- Ronal Bista: *"An EEG-Driven Gameplay Control Pipeline Without Physical Input"*
- Muhammad Ali Asgar Fataymamode: *"Visual Data Exploration through LLM-Guided Query Selection"*
- Yaghyesh Ghimire: *"Web Table Understanding: Column Type Annotation"*
- Rohan Sabu: *"WikiEntities: Entity and Predicate Linking in Wikidata Using LLM-Augmented Vector Matching"*
- Haad Mehboob: *"GNNs for Predicting Neighborhood-Level Population Growth in Abu Dhabi Using Geospatial Data"*
- Harsh Sahu: *"Enhancing JAR Multi-Table Retrieval with LLM-Guided Compatibility and Data-Aware Embeddings"*
- Nathan Behailu: *"Robust GNN Explanations via Causally Informed Policy Optimization"*

AY 2024 – 2025

- Ali Abdullah Jafri: *"Predicting Formula 1 Race Outcomes: A Machine Learning Approach"*
- Kristina Aslanian: *"Mapping the UAE's Cultural Heritage Using Computational Text Analysis"*
- Sasha Malik: *"Leveraging Tree of Thoughts Strategy with Large Language Models for Question Answering on Structured Data"*
- Janindu Nanayakkara: *"Kosmograph: Knowledge Graph Visualization Tool"*

AY 2023 – 2024

- Dariga Shokayeva: *"Named-Entity Linking: Mapping Column Names to Wikidata Properties"*
- Sagynbek Talgatuly: *"VocabBooster: A Personalized English Vocabulary Learning App Powered by Machine Learning and Taxonomies"*

AY 2022 – 2023

- Ryoji Kubo: “*Learning Explanations for Graph Neural Networks based Knowledge Graph Tasks*”
- Kartikey Singhal: “*Multi-Modal Content Retrieval in Wikipedia*”
- Yernar Mukayev: “*University orientated knowledge platform using crowdsourcing for summarization*”
- Hanzalla Usman: “*Kariger: A Crowd Sourcing Web-App for Physical Tasks in the Gig Economy*”

AY 2021 – 2022

- Fuseini Sunnuma: “*WikiContribute: A Light-weight Context-aware Mobile Application for Editing and Browsing Wikidata*”
- Wajahat Mirza: “*Wikidata2Vec Embeddings & Systematization of research on Entity Enrichment of Transformer based Language Models*”
- Prabhav Arora: “*An evaluation of state of the art Neural Architectures utilized for Time Series Forecasting*”
- Seif Kassem: “*Reinforcement Learning Framework for Data Cleaning*”
- Omar Hussein: “*Building a Tool to Assess the Quality of Writing in Wikipedia Articles*”

AY 2020 – 2021

- Branden Kang: “*Link Predictions Using PyTorch-BigGraph and Gensim*”
- Estelle Ocran: “*Learning Entity Embeddings from Wikipedia and Wikidata for Entity or Article Prediction*”

Ph.D. Pathways Program Mentoring at NYU Abu Dhabi

- Salama Al-Ali. Started Summer 2026.

Master’s Capstone Advising at NYU New York

- Shuang Gao: “*Hyperbolic Question Answering over Knowledge Graphs*” – Fall 2020

Dissertation Committees

- Abdelouahab Khelifati, Ph.D. student, University of Fribourg, Switzerland (2025)
Thesis: “*A Holistic Approach for Time Series Management: Unifying Data Storage and Data Processing*”
- Natalia Ostapuk, Ph.D. student, University of Fribourg, Switzerland (2024)
Thesis: “*Enhancing Structured Knowledge Representation in Large-Scale Data Environments*”

Qualifying Exam Committees

- Anh Mai, NYU Courant. 2024

Master’s and Bachelor’s Student Projects at the University of Fribourg

- Alexander Striffeler: “*Benchmarking Big Data Infrastructures*” – 2015
- Marwa Bouzeyane: “*Grammatical Errors Detection Based On N-gram Analysis*” – 2015
- Stefan Nüesch: “*Real-Time Anomaly Detection in Water Distribution Networks using Spark Streaming*” – 2014
- Phokham Nonava: “*HDFS Blocks Placement Strategy*” – 2014
- Simpal Kumar: “*Real Time Data Analysis for Water Distribution Network using Storm*” – 2014
- Dani Rotzetter: “*Crowd-Flow Designer: An Open-Source Toolkit to Design and Run Complex Crowd-Sourced Tasks*” – 2014
- Victor Felder: “*Openturk: An Implementation of the Pick-A-Crowd Architecture*” – 2013

PRESENTATIONS AND TUTORIALS

Presentations

- *WikiRAG: Revisiting Wikidata KGC Datasets with Community Updates and Retrieval-Augmented Generation*, KDD 2025 – Datasets and Benchmarks presentation.
- *RAW-Explainer: Post-hoc Explanations of Graph Neural Networks on Knowledge Graphs*, MLoG-GenAI workshop at KDD 2025 (non-proceedings) – poster presentation.
- *Knowledge Discovery and Exploration with LLMs and Graph ML*, AIE Conference 2025 (NYU Abu Dhabi) – invited talk.
- *Machine-in-the-loop: A New Paradigm of Crowdsourcing for Wikipedia Editing*, WSDM 2023 Crowd Science Workshop (Singapore) – invited talk.
- *Deadline-Aware Fair Scheduling for Multi-Tenant Crowd-Powered Systems*, CSCW 2019 (Texas, USA) – invited presentation.
- *Crowd Size Estimation*, Moore-Sloan Data Science Summit 2018 (Utah, USA).
- *Big Data for Executives*, IIMT guest lecture (Fribourg, Switzerland, 2016).
- *Quality and Performance Optimizations in Microtask Crowdsourcing*, MIT CSAIL (Boston, MA, 2014).
- *Quality and Performance Optimizations in Microtask Crowdsourcing*, CMU (Pittsburgh, PA, 2014).
- *Extending the OLTP-Bench Framework for Big Data Systems*, Workshop on Big Data Benchmarking 2014 (Potsdam, Germany).
- *Mechanical Cheat: Spamming Schemes and Adversarial Techniques on Crowdsourcing Platforms*, Crowd-Search 2012 workshop at WWW (Lyon, France).

Conference Tutorials

- *Aggregation Techniques in Crowdsourcing: Multiple Choice Questions and Beyond* CIKM 2021 (online) – Organizers: Djellel Difallah, Alessandro Checco.
- *Using Crowdsourcing Effectively for Social Media Research* ICWSM 2016 (Cologne, Germany) – Organizers: Ujwal Gadiraju, Gianluca Demartini, Djellel Difallah, Michele Catasta.
- *It's Getting Crowded! How to Use Crowdsourcing Effectively for Web Science Research* Web Science 2016 (Hannover, Germany) – Organizers: Ujwal Gadiraju, Gianluca Demartini, Djellel Difallah, Michele Catasta.

SERVICE

University Service – at NYU Abu Dhabi

- Computer Science Program Head (Sep 2025 – Present)
- Minor in AI and Machine Learning – Proposal development (2026)
- Science and Engineering Curriculum Committee (CS representative) (Sep 2022 – 2025)
- Undergraduate Curriculum Committee (At-large member) (Sep 2021 – Dec 2023)
- Master of Science in Interdisciplinary Data Science and Artificial Intelligence – Proposal working group member (2021 – 2022)
- Search Committee Member (CS Professor, Open Rank) (AY 2021 – 2022)
- Search Committee Member (CS Instructor) (AY 2021 – 2022)
- Contract Faculty Review Committees (Chair of one committee)
- Undergraduate Research Grants Reviewer (CS proposals) (2021 – Present)
- Piano Prize Committee Member (2022 – 2023)

Senior Program Committee

- ECIR (SPC): 2027
- ICDM (Area Chair): 2026, 2024, 2023
- CIKM (Research Track SPC): 2026, 2025, 2024, 2023
- CIKM (Applied Research Track SPC): 2022, 2021, 2020
- HCOMP (SPC): 2024
- AAAI: 2024, 2023

Program Committee

- IEEE International Conference on Data Mining (ICDM): 2025, 2022, 2021, 2020, 2019, 2018
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD): 2025, 2024, 2023, 2022, 2021, 2020, 2019, 2018, 2017
- ACM International Conference on Information and Knowledge Management (CIKM): 2025, 2024, 2023, 2022, 2021, 2020, 2018, 2017
- International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR): 2026, 2025, 2024, 2023, 2022, 2021, 2020
- ACM International Conference on Web Search and Data Mining (WSDM): 2025, 2024, 2023, 2022, 2021
- ACM Web Conference (WWW): 2023, 2022, 2021, 2020, 2019, 2018, 2017
- AAAI Conference on Artificial Intelligence (AAAI): 2022, 2021, 2020
- IEEE International Conference on Data Engineering (ICDE): 2026, 2025
- AAAI Conference on Human Computation and Crowdsourcing (HCOMP): 2024, 2022, 2021, 2020, 2019
- International Semantic Web Conference (ISWC): 2022, 2021, 2020, 2019
- SIAM International Conference on Data Mining (SDM): 2022
- Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD): 2025
- ACM International Conference on Management of Data (SIGMOD): 2025
- International AAAI Conference on Web and Social Media (ICWSM): 2018
- International Conference on Web Engineering (ICWE): 2016
- International Conference on Social Informatics (SocInfo): 2022, 2020, 2019
- Wiki Workshop: 2023, 2022, 2021
- Wiki NLP Workshop: 2025

Conference Session Chair

- Session Chair: ICDM 2024, ISWC 2022, SIGIR 2021, KDD 2021